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Physics-Infused Reduced-Order Modeling of Hypersonic Systems

An Overview of Recent Advances and Future Directions

Carlos A. Vargas Venegas

April 2, 2026

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SAND NO. 2014-00000

VIRTUAL TWINS OF HYPERSONIC SYSTEMS

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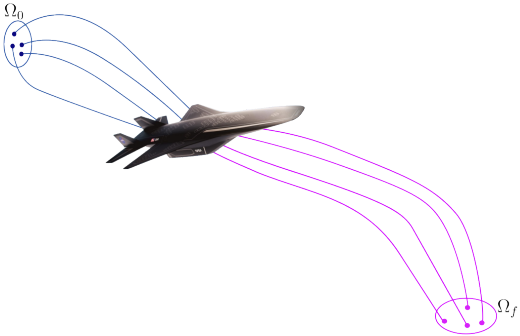

$\mathbf{u}$: high-dimensional states, $\mathbf{F}$: non-linear dynamics $\mathbf{p}$: design/control parameters, $\mathbf{z}_{\mathrm{HF}}$: high-fidelity outputs, $\mathcal{J}$: cost functional, l: running cost, $\mathbf{Q}$: safety bound, $\mathbf{g}$: constraint function

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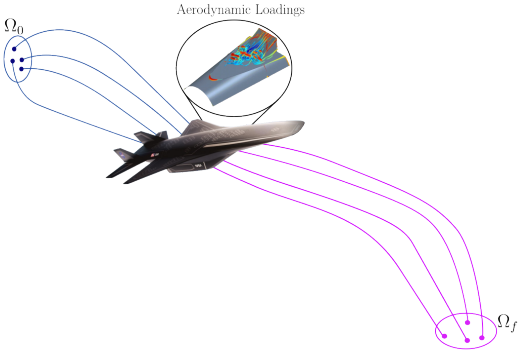

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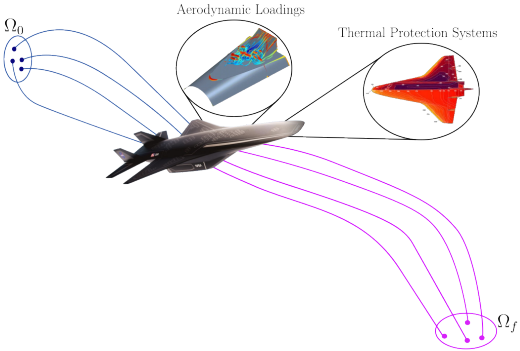

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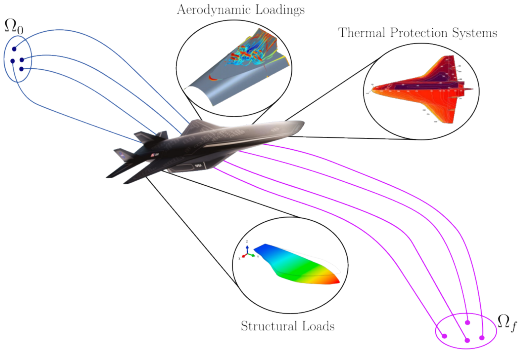

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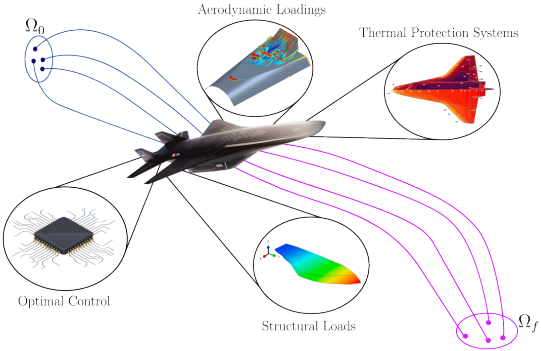

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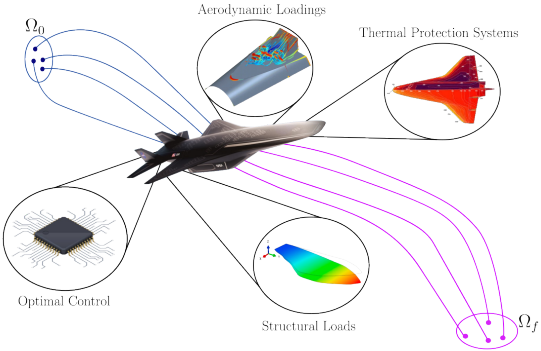
Finite-Element Model (FEM)



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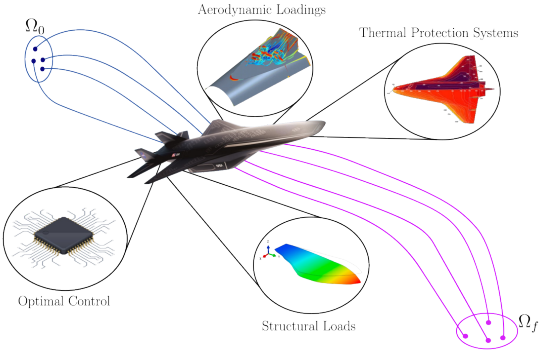

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$$\dot{\mathbf{u}} = \mathbf{F}\left(\mathbf{u}; \mathbf{p}\right)$$

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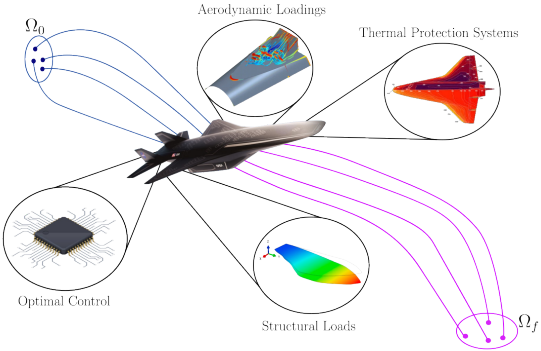

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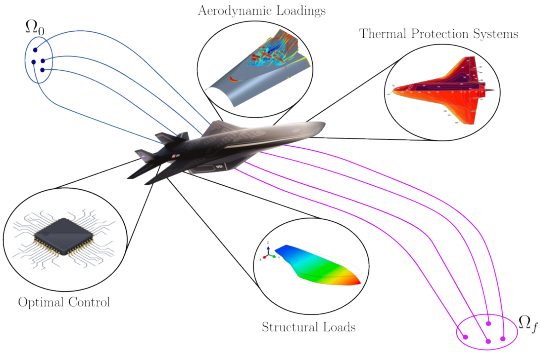

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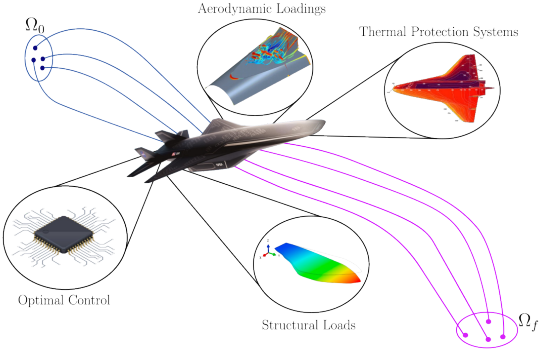

Optimal Control Problem (OCP)

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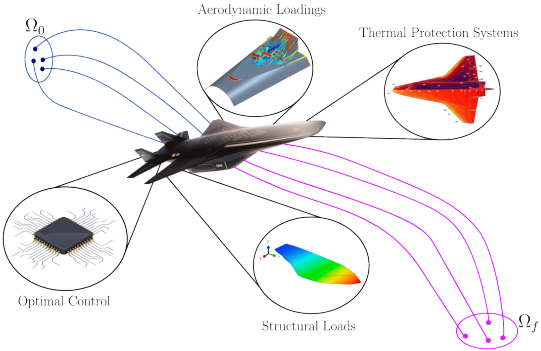

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$$\min_{\mathbf{p}} \mathcal{J} = \int_{t_0}^{t_f} \ell\left(\mathbf{u}, \mathbf{p}\right) dt$$

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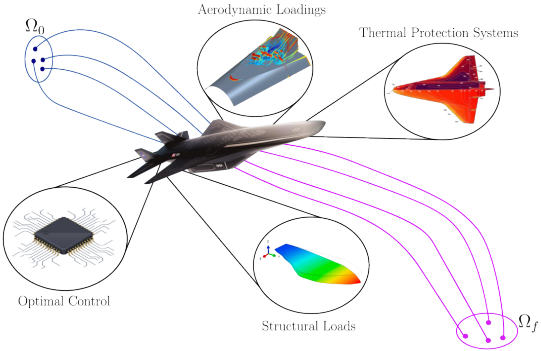

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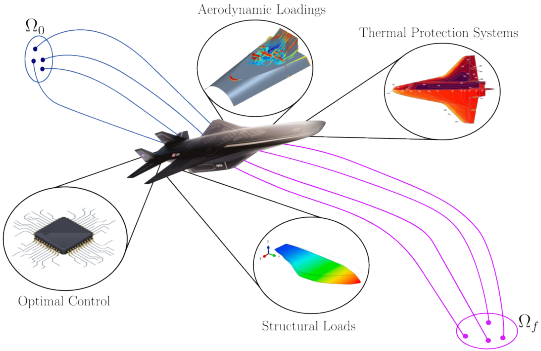

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$$\mathbf{Q} \geq \mathbf{g}\left(\mathbf{u}(t), t; \mathbf{p}\right)$$

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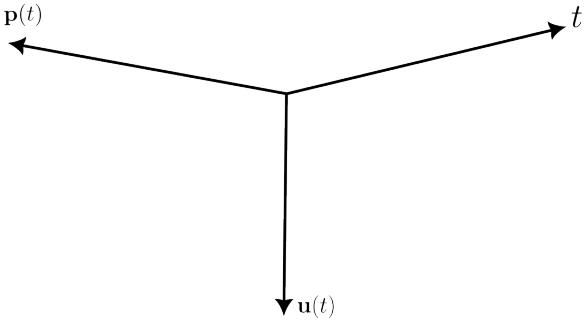
EXPLORATION OF DESIGN SPACE IS COSTLY

States	Parameters	Discrete Time Steps
$\mathbf{u}(t) \in \mathbb{R}^{n_u}, (n_u = 10^2 - 10^6)$	$\mathbf{p}(t) \in \mathbb{R}^{n_p}, (n_p = 10^1 - 10^3)$	$t \in \mathbb{R}^{n_t}, (n_t = 10^1 - 10^2)$



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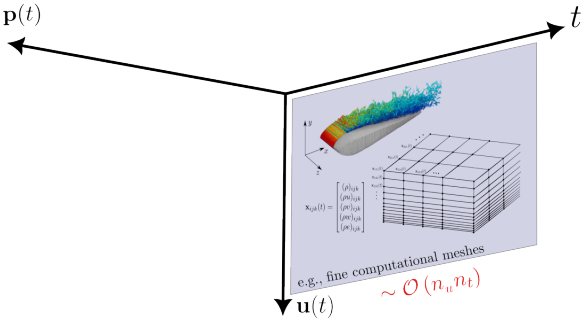

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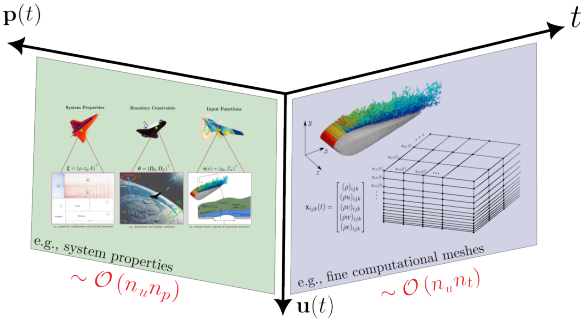

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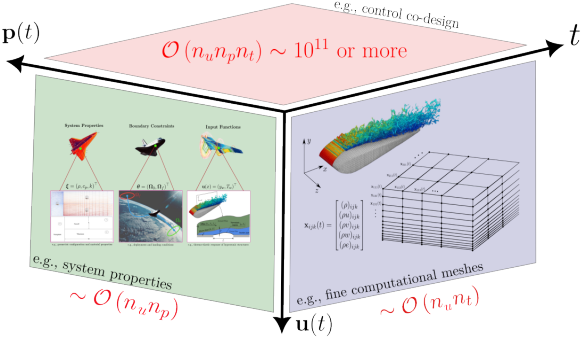

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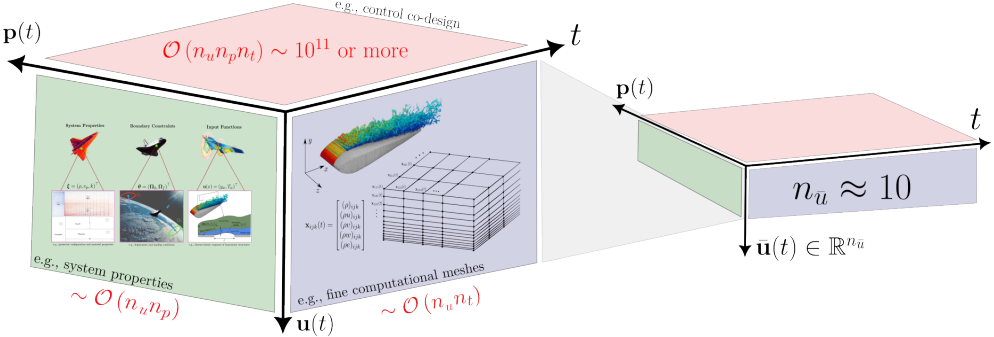

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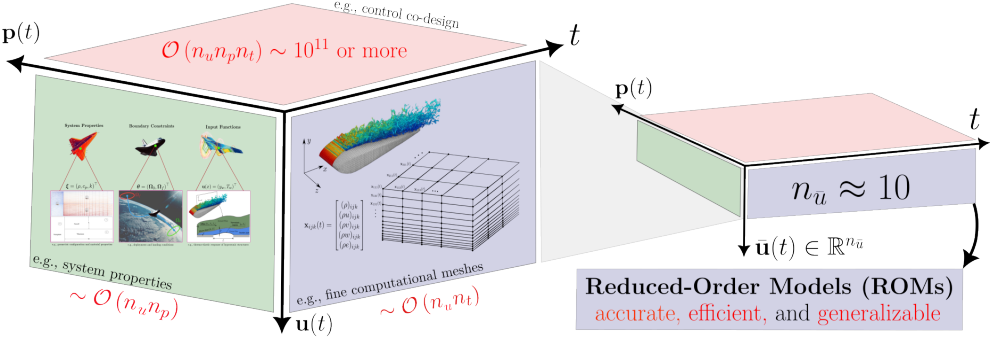

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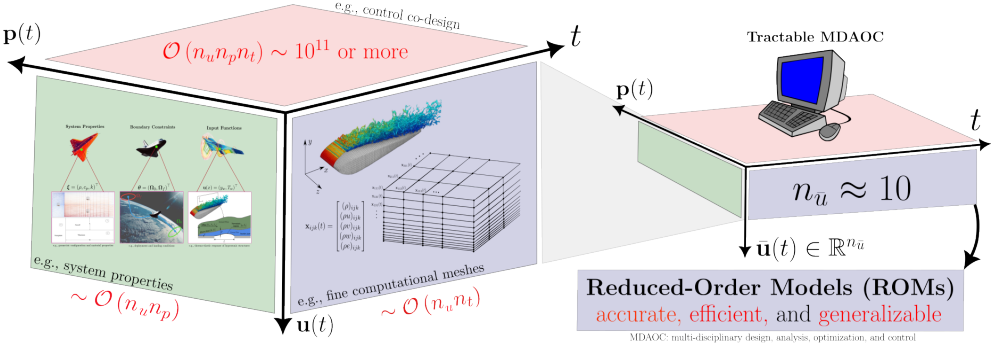

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